

Name: _____

In-class Quiz #07: GPIO

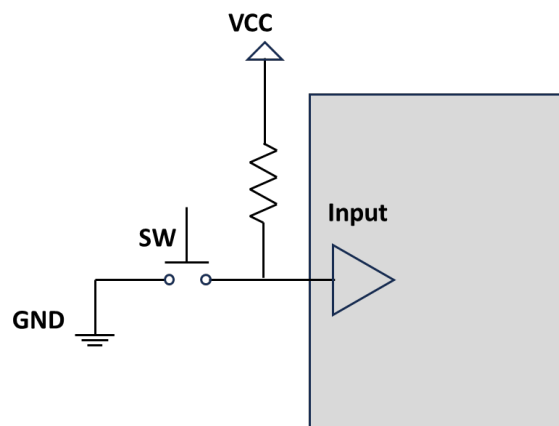
Note: All the codes are assumed to run on the TM4C123G microcontroller (TI LaunchPad).

1. What is the hexadecimal bit-specific address for bits 0-4 of Port B's GPIO Data (GPIODATA) register? (Assume APB.)

Answer: 0x_____

2. Assume V_{cc} in the image is equal to 3.3 [V]. What is the voltage at the input pin when the switch is closed (the button is pressed?) Circle the answer.

Answer: 3.3 V 0V



3. Which port clocks are activated by executing the following code? (Circle all that apply.)

PORTS: A B C D E F

```
LDR R1, =SYSCTL_RCGCGPIO_R
LDR R0, [R1]
ORR R0, R0, #0x1A
STR R0, [R1]
NOP
```

4. Which pins on Port B are configured to be analog pins after executing the code below?

(Circle all that apply.)

PINS: 0 1 2 3 4 5 6 7

```
LDR  R1, =GPIO_PORTB_AMSEL_R
LDR  R0, [R1]
ORR  R0, R0, #0xE8
BIC  R0, R0, #0x17
STR  R0, [R1]
NOP
```

5. What is the hexadecimal address for **Port A's** Alternative Function Select (GPIOAFSEL) register? (Assume APB.)

Answer: 0x_____

6. Write the code to configure bit 0-3 for Port B as output and bits 4-7 as inputs. You are not allowed to change bits 8-31 of the configuration register. (Do not use labels in your code. Use the actual address for the configuration register.)
7. Write the code to enable the pull-down register on pins 0-3 on Port C (APB bus). Do not change the values for any other pins besides 0-3. (Do not use labels in your code. Use the actual address for the configuration register.)
8. What is the base address for the Port D GPIO Data (GPIODATA) register? (Assume APB.)

Address: 0x_____

9. Assume that initially all LEDs on Port F of the LaunchPad are off. After the code below is executed, what is the state of the three built-in LEDs connected to the Port F? (Circle the correct answers.)

Red LED is: on off
Blue LED is: on off
Green LED is: on off

```
LDR  R1, =0x400253FC
MOV  R0, #2_00010
STR  R0, [R1]
```

10. After executing the code below, the value in R0 is 0x00000000. What do you know about the state of the switches/buttons on the LaunchPad?

Switch 1 (SW1 = PortF pin 4) is ...	pressed	not pressed	don't know
Switch 2 (SW2 = PortF pin 0) is ...	pressed	not pressed	don't know

```
LDR  R1, =0x40025004
LDR  R0, [R1]
```